MATH MA’AM ACADEMY

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STANDARD VI • SCIENCE

UNIT 4: AIR & OXYGEN

Comprehensive revision notes based on the supplied textbook pages, with essential explanations and clarifications.

QUICK REVISION

Topic	Key facts
Dry air composition	Nitrogen 78.08%; oxygen 20.95%; argon 0.93%; carbon dioxide about 0.036% (textbook figures).
Atmosphere	About 99% of its mass lies within 32 km of Earth's surface.
Lowest layer	Troposphere; most weather occurs here.
Ozone layer	Mainly in the stratosphere; absorbs harmful ultraviolet radiation.
Acid rain	Mainly associated with sulphur dioxide and nitrogen oxides.
Oxygen test	Glowing splint or burning matchstick burns more brightly.
Carbon monoxide	Combines with haemoglobin to form carboxyhaemoglobin.

Atmospheric composition values are rounded textbook estimates. Water vapour varies and is not included in the dry-air percentages.

1. THE ATMOSPHERE

The atmosphere is the envelope or blanket of air that completely surrounds Earth. It is held around the planet by Earth's gravitational pull.

- The atmosphere supplies oxygen needed by most living organisms and helps maintain conditions suitable for life.
- It absorbs some harmful solar radiation and causes many small meteoroids to burn up.
- The textbook describes the atmosphere as extending to about 1,000 km.
- About 99% of its total mass is concentrated within the first 32 km because gravity keeps most air close to Earth's surface.

2. COMPOSITION OF THE ATMOSPHERE

Air is a mixture of gases and tiny solid particles. Its composition varies with time and location, particularly because water vapour and dust vary. Clean, dry air has a relatively uniform composition.

Component	Textbook value	Importance
Nitrogen (N_2)	78.08%	Most abundant gas in air.
Oxygen (O_2)	20.95%	Required for respiration; supports combustion.
Argon (Ar)	0.93%	An inert noble gas.
Carbon dioxide (CO_2)	0.036%	Used by plants in photosynthesis; greenhouse gas.
Neon (Ne)	0.002%	Noble gas used in illuminated signs.

- Water vapour: may vary from almost zero to about 4% by volume in the textbook account. It is important for clouds, rain and weather.
- Dust: varies by location and provides surfaces on which water vapour can condense.

Remember: the table describes clean, dry air. Water vapour is variable and is not included in dry-air percentages. Carbon dioxide levels have changed since older textbooks were published.

3. LAYERS OF THE ATMOSPHERE

The atmosphere has five main layers. They are identified mainly by the way temperature changes with increasing height.

Layer	Height / temperature trend	Main features
Troposphere	Surface to about 18 km at Equator and 8 km at poles; temperature decreases with height.	Lowest layer; most water vapour, clouds and weather. Upper boundary: tropopause.
Stratosphere	To about 50 km; temperature increases with height.	Contains ozone, which absorbs ultraviolet radiation. Stable conditions; suitable flight region for large jets. Upper boundary: stratopause.
Mesosphere	To about 80 km; temperature decreases, reaching about −90°C in textbook.	Coldest atmospheric layer; many meteoroids burn up here. Upper boundary: mesopause.
Thermosphere	Above mesosphere; textbook describes rise beginning around 90 km.	Absorbs X-rays and shortwave UV; temperatures rise sharply. Includes much of ionosphere.
Exosphere	Outermost region; textbook gives extent to about 960 km, merging with space.	Extremely thin gases; particles can escape into space.

Important features

- Troposphere: 'tropo' means change. Nearly all ordinary weather occurs here. Temperature at the tropopause may be about −58°C in the textbook.
- Stratosphere: ozone absorbs ultraviolet radiation, causing temperature to increase with height. It is relatively free from clouds and weather disturbances.
- Mesosphere: temperature falls with height; its upper boundary is the mesopause.
- Thermosphere: high-energy solar radiation ionises gas particles. Electrically charged particles are called ions.
- Ionosphere: charged gases affect/reflect radio waves and help long-distance radio communication.
- Exosphere: the outermost, very thin region, gradually merging with interplanetary space.

Temperature trend: Troposphere decreases → Stratosphere increases → Mesosphere decreases → Thermosphere increases.

Exact layer heights and temperatures vary with latitude, season and solar activity. Learn the textbook values for the prescribed examination.

4. USES OF NITROGEN, OXYGEN AND CARBON DIOXIDE

Gas	Uses and importance
Nitrogen	• Dilutes oxygen in air and slows combustion. • Needed for plant growth and protein formation; plants absorb nitrogen compounds such as nitrates and ammonium salts. • Used in manufacture of ammonia. • Used in food packaging to reduce oxidation and spoilage.
Oxygen	• Essential for respiration. • Medical oxygen is supplied when required. • Liquid oxygen is an oxidiser in rocket systems. • Supports high-temperature flames for welding and cutting. • Used in some water and sewage treatment processes.
Carbon dioxide	• Used by green plants in photosynthesis. • Dissolved in water for carbonated drinks. • Used in some fire extinguishers. • Solid CO_2, called dry ice, is used as a refrigerant.

5. USES OF NOBLE GASES

Gas	Common uses
Argon	Filling electric bulbs; shielding gas in some welding; specialised fire-suppression systems.
Helium	Balloons and airships because it is light and non-flammable; cryogenics for very low temperatures; meteorological uses.
Neon	Neon sign tubes and signboards; indicator lamps; specialised lighting and lasers.

- Noble gases have very low chemical reactivity under ordinary conditions.
- Helium is safer than hydrogen for balloons because helium is non-flammable whereas hydrogen is highly flammable.

6. AIR POLLUTION

Air pollution is contamination of air by harmful substances such as dust, smoke and poisonous gases. These harmful substances are called pollutants.

Pollutant	Important information
Carbon monoxide (CO)	Poisonous gas produced by incomplete combustion of fuels.
Carbon dioxide (CO_2)	Produced by combustion and respiration; excess atmospheric CO_2 contributes to global warming.
Sulphur dioxide (SO_2)	Released by burning sulphur-containing fuels and some industries; contributes to acid rain.
Nitrogen oxides (NO, NO_2)	Produced during high-temperature combustion, including vehicle engines; contribute to smog and acid rain.
Ozone (O_3)	Beneficial in stratosphere, but harmful to breathe near ground level at high concentrations.
Hydrogen sulphide (H_2S)	Toxic gas released by some industrial processes and decomposition.
Dust and smoke	Particles that irritate the respiratory system and reduce air quality.

Sources of air pollution

- Fuel burning: coal, wood, kerosene, petrol and diesel release smoke, carbon particles, CO, CO_2 and sometimes SO_2.
- Automobiles: cars, buses, trucks and two-wheelers release exhaust gases. Vehicle exhaust is a major urban source.
- Industries: factories may release chlorine, SO_2, NO_2, CO_2 and other pollutants, depending on the process.
- Smoking: tobacco smoke contains carbon monoxide, nicotine and other harmful chemicals.
- Dust: traffic, construction and industrial operations release dust particles.

Effects of air pollutants

- Can cause coughing, throat irritation, breathlessness, bronchitis, eye irritation and allergies.
- Long-term exposure to some pollutants increases the risk of serious respiratory and cardiovascular diseases.
- Ground-level ozone irritates the nose, throat and lungs and can damage rubber and plants.
- Carbon monoxide: combines strongly with haemoglobin to form carboxyhaemoglobin. This reduces the blood's oxygen-carrying capacity and can cause headache, dizziness, suffocation and death at high exposure.
- Dust and smoke irritate the eyes and respiratory tract and can worsen existing lung conditions.

7. PREVENTION OF AIR POLLUTION

- Plant and protect trees, especially in urban areas.
- Use cleaner fuels such as natural gas where appropriate instead of more polluting fuels.
- Prefer cleaner energy sources such as solar, wind, tidal and low-emission electrical energy.
- Maintain vehicles, use public transport, walk or cycle when practical, and avoid unnecessary idling.
- Install and maintain pollution-control equipment such as electrostatic precipitators in industrial chimneys.
- Control industrial emissions and follow prescribed emission standards.

Electrostatic precipitator

An electrostatic precipitator removes fine particles from industrial exhaust. Particles are electrically charged and attracted to oppositely charged collecting plates, reducing particulate emissions.

8. ACID RAIN

Acid rain is precipitation that becomes unusually acidic because certain air pollutants dissolve in atmospheric moisture and form acids. The main gases involved are sulphur dioxide and nitrogen oxides.

- Sulphur dioxide can form sulphurous acid and, after further oxidation, sulphuric acid.
- Nitrogen oxides can form nitric acid in the atmosphere.
- Carbon dioxide dissolves in rainwater to form weak carbonic acid naturally; it is not the main cause of damaging acid rain.

Harmful effects of acid rain

- Makes lakes and rivers more acidic, harming fish and other aquatic organisms.
- Damages leaves and living tissues and can affect forests and crops.
- Erodes limestone, marble and cement-based structures and damages monuments and old buildings.
- Corrodes steel and other metals, shortening the life of structures.

Prevention and control

- Reduce SO_2 and nitrogen oxide emissions by using cleaner fuels and industrial processes.
- Use emission-control systems in factories and vehicles.
- Improve energy efficiency and reduce unnecessary fossil-fuel combustion.

9. OXYGEN: DISCOVERY AND PREPARATION

Oxygen is essential for respiration and supports combustion. The textbook credits Swedish chemist Karl Scheele with the first preparation of oxygen. Joseph Priestley also prepared oxygen by heating red oxide of mercury. Antoine Lavoisier later named the gas oxygen and explained its role in combustion.

Laboratory preparation of oxygen

- Oxygen can be prepared by heating certain oxygen-rich compounds, such as potassium permanganate or potassium chlorate, under controlled laboratory conditions.
- Hydrogen peroxide decomposes to produce water and oxygen; manganese dioxide can act as a catalyst.
- A catalyst changes the rate of a reaction without being consumed in the overall reaction.

Method	Simplified reaction / role
Hydrogen peroxide decomposition	$2H_2O_2 \rightarrow 2H_2O + O_2$; MnO_2 can act as a catalyst.
Heating potassium chlorate	$2KClO_3 \rightarrow 2KCl + 3O_2$; MnO_2 is commonly used as a catalyst.
Heating potassium permanganate	On heating, potassium permanganate decomposes and releases oxygen along with solid products.

10. PROPERTIES OF OXYGEN

Property / test	Observation
Colour	Colourless
Odour	Odourless
Effect on litmus	No change to red or blue litmus; neutral in this test.
Solubility	Slightly soluble in water.
Density	Slightly heavier than air.
Effect on burning material	A glowing splint or burning matchstick burns more brightly.

Confirmatory test for oxygen

A glowing splint introduced into a sample of oxygen relights or burns more brightly. Oxygen itself does not burn; it supports combustion.

11. IMPORTANT COMPARISONS

Feature	Oxygen	Carbon dioxide
Role in combustion	Supports combustion; is not itself a fuel.	Does not support ordinary combustion; used in some fire extinguishers.
Role in living systems	Required for aerobic respiration.	Used by plants in photosynthesis; high concentrations can harm humans.
Test	Glowing splint relights or burns more brightly.	Turns limewater milky due to calcium carbonate formation.

KEY TERMS

Term	Meaning
Atmosphere	Envelope of gases surrounding Earth.
Troposphere	Lowest atmospheric layer where most weather occurs.
Tropopause	Boundary between troposphere and stratosphere.
Stratosphere	Layer containing the ozone-rich region; temperature generally rises with height.
Ozone layer	Region with relatively high ozone concentration that absorbs much harmful UV radiation.
Mesosphere	Layer above stratosphere where temperature decreases with height.
Thermosphere	Upper layer where absorption of high-energy solar radiation causes temperature to rise.
Ionisation	Formation of electrically charged particles (ions).
Ionosphere	Region containing ions that affect radio-wave propagation.
Air pollution	Contamination of air by harmful gases, smoke and particles.
Acid rain	Unusually acidic precipitation formed when certain air pollutants dissolve in atmospheric moisture.
Catalyst	Substance that changes reaction rate without being consumed overall.

Revise gas percentages, layer sequence and temperature trends, uses of gases, pollution sources and effects, prevention methods, acid rain and oxygen laboratory observations.